LEFT DUALS IN THE MONOIDAL CATEGORY OF COMPLEXES ARE DEGREEWISE LEFT DUALS WITH A SIGN-TWISTED DIFFERENTIAL

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement span contains the author's displayed formula for the differential together with its three-part enumerated list and the converse construction, all delivered verbatim. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Dual pairs in a monoidal category and the two triangle identities. invoked at proof lines 18494-18499. Are unwound into the degreewise dual identities of part (3).
- Left duals of modules and the identification of the dual module with Hom_R(M, R). invoked at proof lines 18500 and 18509. Gives part (2) and the identification used in the formula for the differential.
- The sign rule for the differential of a total complex. used at proof lines 18505-18509. Is what produces the factor -(-1)^n in the differential of the dual complex.

Source reference. Original lemma 0FFP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19294–19294. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 18419: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18419–18419. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $M^\bullet$ be a complex of R-modules. Let $N^\bullet, \eta, \epsilon$ be a left dual of $M^\bullet$ in the monoidal category of complexes of R-modules. Then*

(1) *$M^\bullet$ and $N^\bullet$ are bounded,*
(2) *M^n and N^n are finite projective R-modules,*

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(3) *writing* $\epsilon = \sum \epsilon_n$ *with* $\epsilon_n : N^{-n} \otimes_R M^n \to R$ *and* $\eta = \sum \eta_n$ *with* $\eta_n : R \to M^n \otimes_R N^{-n}$ *then* $(N^{-n}, \eta_n, \epsilon_n)$ *is the left dual of* M^n *as in Lemma 18419,*
(4) *the differential* $d_N^n : N^n \to N^{n+1}$ *is equal to* $-(-1)^n$ *times the map*

$$N^n = \mathrm{Hom}_R(M^{-n}, R) \xrightarrow{d_M^{-n-1}} \mathrm{Hom}_R(M^{-n-1}, R) = N^{n+1}$$

where the equality signs are the identifications from Lemma 18419 part (2).

Conversely, given a bounded complex $M^\bullet$ *of finite projective* R*-modules, setting* $N^n = \mathrm{Hom}_R(M^{-n}, R)$ *with differentials as above, setting* $\epsilon = \sum \epsilon_n$ *with* $\epsilon_n : N^{-n} \otimes_R M^n \to R$ *given by evaluation, and setting* $\eta = \sum \eta_n$ *with* $\eta_n : R \to M^n \otimes_R N^{-n}$ *mapping* 1 *to* id_{M_n} *we obtain a left dual of* $M^\bullet$ *in the monoidal category of complexes of* R*-modules.*

Proof. Since $(1 \otimes \epsilon) \circ (\eta \otimes 1) = \mathrm{id}_{M^\bullet}$ and $(\epsilon \otimes 1) \circ (1 \otimes \eta) = \mathrm{id}_{N^\bullet}$ by Categories, Definition 0FFP we see immediately that we have $(1 \otimes \epsilon_n) \circ (\eta_n \otimes 1) = \mathrm{id}_{M^n}$ and $(\epsilon_n \otimes 1) \circ (1 \otimes \eta_n) = \mathrm{id}_{N^{-n}}$ which proves (3). By Lemma 18419 we have (2). Since the sum $\eta = \sum \eta_n$ is finite, we get (1). Since $\eta = \sum \eta_n$ is a map of complexes $R \to \mathrm{Tot}(M^\bullet \otimes_R N^\bullet)$ we see that

$$(d_M^{-n-1} \otimes 1) \circ \eta_{-n-1} + (-1)^n (1 \otimes d_N^{-n}) \circ \eta_{-n} = 0$$

by our choice of signs for the differential on $\mathrm{Tot}(M^\bullet \otimes_R N^\bullet)$. Unwinding definitions, this proves (4). To see the final statement of the lemma one reads the above backwards. □